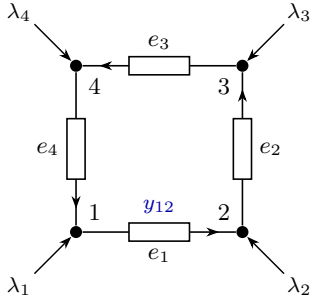
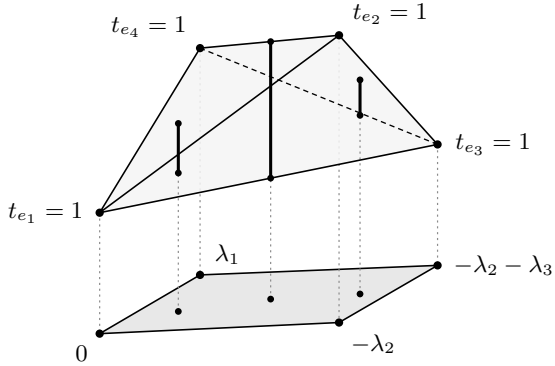
(a) The square ring Γ



$|\Gamma_1| = 2|\Gamma_0| - 4 = 4, \quad h_1(\Gamma_1) = 1$

(b) Resistance ratios over the loop current



$|\Gamma_1| - 1 = 3$ resistance ratios

$[t] \mapsto y_{12}(t) = \dfrac{\sum_m t_{e_m} Y_m}{\sum_m t_{e_m}}$

$2h_1(\Gamma_1) = 2$ components of y_{12}

fibres: $3 - 2 = 1$ ratio direction that moves no current

$e^{-\sum_e z_e^{+\dot{a}} y_{e,+\dot{a}}} \left(-d^2 y_{12}\right)$: top form on the y_{12}-plane, pulled back along $[t] \mapsto y_{12}(t) \Rightarrow$ closed